

Statistics

Questions and Answers:

Q1 : A die is rolled. What is the probability of getting:

(a) An even number

(b) A number greater than 4

Ans:

{1, 2, 3, 4, 5, 6} (total = 6 outcomes)

(a) An even number

Even numbers on a die: {2, 4, 6} → 3 outcomes

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

(b) Probability of getting a number greater than 4

Numbers greater than 4: {5, 6} → 2 outcomes

$$P(\text{greater than 4}) = \frac{2}{6} = \frac{1}{3}$$

Final Answers:

- (a) $\frac{1}{2}$
- (b) $\frac{1}{3}$

Q2 : In a class of 50 students:

20 like Mathematics (M)

15 like Science (S)

5 like both subjects

What is the probability that a student chosen at random likes Mathematics or Science?

Ans:

We use the formula:

$$P(M \cup S) = \frac{n(M) + n(S) - n(M \cap S)}{\text{Total students}}$$

Substitute the values:

$$P(M \cup S) = \frac{20 + 15 - 5}{50} = \frac{30}{50} = \frac{3}{5}$$

Final Answer:

$$\boxed{\frac{3}{5}} \text{ or } 0.6$$

Q3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

Ans:

Since the first ball drawn is red, we update the situation:

- Initially: 3 red, 2 blue (total 5)
- After drawing 1 red (without replacement):
Remaining = 2 red, 2 blue (total 4)
- Probability that the next ball is also red:**

$$P(\text{second red}) = \frac{\text{remaining red balls}}{\text{total remaining balls}} = \frac{2}{4} = \frac{1}{2}$$

Final Answer:

$$\boxed{\frac{1}{2}}$$

Q4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

Ans:

Stratified Sampling.

Why?

The population is uneven (60% boys, 40% girls), but you want equal representation in your sample.

- In Simple Random Sampling, selection is purely random, so your sample will likely reflect the same ratio (more boys than girls). You may not get equal numbers.
- In Stratified Sampling, you divide the population into groups (called strata) — here, boys and girls — and then select equal numbers from each group.

Q5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

Ans:

Sampling Error Formula:

$$\text{Sampling Error} = \text{Sample Mean} - \text{Population Mean}$$

Substitute the values:

$$= 158 - 160 = -2$$

Final Answer:

$$\boxed{-2 \text{ cm}}$$

Q6: The population mean salary is ₹50,000 with $\sigma = ₹5,000$. If we take a sample of 100 employees, what is the standard error of the mean (SEM)?

Ans :

Formula:

$$\text{Standard Error of Mean (SEM)} = \frac{\sigma}{\sqrt{n}}$$

Substitute the values:

$$\text{SEM} = \frac{5000}{\sqrt{100}} = \frac{5000}{10} = 500$$

Final Answer:

$$\boxed{₹500}$$

Q7: In a group of 100 students: 40 like Cricket (C) 30 like Football (F) 10 like both Cricket and Football Find the probability that a student likes at least one sport.

Ans:

We use the formula:

$$P(C \cup F) = \frac{n(C) + n(F) - n(C \cap F)}{\text{Total students}}$$

Substitute the values:

$$P(C \cup F) = \frac{40 + 30 - 10}{100} = \frac{60}{100} = \frac{3}{5}$$

Final Answer:

$$\boxed{\frac{3}{5}} \text{ or } 0.6$$

Q8: From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

Ans:

Step 1: Probability first card is an Ace

$$P(\text{first Ace}) = \frac{4}{52}$$

Step 2: Probability second card is an Ace (without replacement)

Now 3 Aces remain out of 51 cards:

$$P(\text{second Ace}) = \frac{3}{51}$$

Step 3: Multiply the probabilities

$$P(\text{both Aces}) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

Final Answer:

$$\boxed{\frac{1}{221}}$$

Q9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Ans:

Defective rate = 2% $\Rightarrow$ Non-defective rate = 98% = 0.98

We want the probability that all 5 bulbs are non-defective:

$$P(\text{all non-defective}) = (0.98)^5$$

Now calculate:

$$(0.98)^5 \approx 0.9039$$

Final Answer:

$$\boxed{0.9039}(\text{approximately})$$

Q10: Differentiate between discrete and continuous random variables with examples.

Ans:

Feature	Discrete Random Variable	Continuous Random Variable
Definition	Takes countable values	Takes uncountable (infinite) values
Nature	Values are separate/distinct	Values are continuous (any value in interval)
Examples of values	0, 1, 2, 3, ...	1.2, 3.75, 5.001, ...
Counting	Can be counted	Cannot be exactly counted, only measured
Probability	Uses probability mass function (PMF)	Uses probability density function (PDF)

EXAMPLE:

Discrete Random Variable

Example: roll a die, possible outcomes are {1, 2, 3, 4, 5, 6} (countable)

Continuous Random Variable

Example: Height can be 160 cm, 160.5 cm, 160.55 cm... (infinite possibilities)